# ATHARVA VINAYAK YADAV

**Firmware & Embedded Systems Engineer (R&D)** | Bare-Metal C/C++ • HIL/ATE Validation • Control Systems

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## PROFESSIONAL SUMMARY

Production-focused R&D Firmware & Embedded Systems Engineer with end-to-end ownership of cyber-physical hardware architectures—bridging bare-metal microcontroller firmware (C/C++, STM32, ESP32), deterministic closed-loop control (100Hz PID, 16-Channel Thermal DAQ, Hardware Watchdogs), and multi-threaded desktop Automated Test Equipment (ATE) / HIL validation suites (Python, PySide6, PyOpenGL). Proven track record of architecting firmware, high-precision capacitance bridge instrumentation (BARC), semiconductor wafer laser controllers, and radiometric LWIR thermography systems.

## TECHNICAL SKILLS

- **Microcontrollers & Bare-Metal Firmware:** C, C++, Embedded C, STM32 (ARM Cortex-M / Black Pill), ESP32, Direct Register Manipulation, Circular DMA, Multi-Channel ADC, Hardware Watchdogs (IWDG/WWDG), Interrupts (ISR/NVIC), Custom Marlin Firmware.
- **Control Theory, Instrumentation & HIL/ATE Validation:** Deterministic 100Hz PID Control Loops, Sub-pF Capacitance Bridge Balancing (BARC), 16-Channel NTC Thermistor DAQ, Active Thermal Gradient Tracking, Fault-State Injection & Safety Cutoffs, Pneumatic Pressure Transducers (MPX5700), Lock-In Phase Detection.
- **Bus Protocols & Hardware Interfacing:** UART, SPI, I2C, DB25 Parallel Hardware Gating, 100 Mbit Ethernet Streaming, Win32 C++ DLL Hardware Interfacing (MarkEzd.dll), Custom Serial Lockstep Protocols (M118), G-Code Kinematics.
- **Supervisory UI, Test Automation & Tooling:** Python, PySide6 / PyQt, PyOpenGL (3D Digital Twins), Multi-Threaded Producer-Consumer ATE Suites, Git, React.js, Node.js.

## PROFESSIONAL EXPERIENCE

- **Specialise Products Pvt. Ltd. (SPPL)** — Mumbai, India **Jan 2026 – Present**

***Firmware Developer — R&D (IoT & Embedded Systems) (May 2026 – Present) | Graduate Engineer Trainee (Jan 2026 – May 2026 | Fast-Track Promoted)***

- **High-Precision Capacitance & Impedance Bridge Instrumentation (BARC R&D Project):** Designed and engineered embedded firmware and ultra-low-noise Analog Front-End (AFE) acquisition pipelines for a high-precision capacitance measurement bridge developed for Bhabha Atomic Research Centre (BARC) standards. Implemented lock-in phase-sensitive detection, high-resolution ADC signal conditioning to resolve sub-picofarad variations with minimal thermal drift/EMI, and automated bridge-balancing calibration over a Python (PySide6) supervisory interface.
- **Automated 16-Channel Thermal DAQ & Safety Watchdog Architecture (SVNIT System):** Engineered bare-metal STM32 C++ firmware for a 16-channel real-time thermistor data acquisition (DAQ) pipeline utilizing non-blocking ADC + Circular DMA transfers. Implemented active thermal gradient tracking (dT/dt), hardware-level emergency cutoff watchdogs to prevent thermal runaway, and a multi-threaded PySide6 ATE desktop validation suite for real-time fault logging.

- **Radiometric LWIR Thermography & Lock-In NDT Inspection Systems (SP2L-TC / SP2L-LTWIS / SP2L-ALESTM):**

  Developed real-time DAQ and 100 Mbit Ethernet packet streaming pipelines for radiometric Long-Wave Infrared (LWIR, 8–14 µm) thermography cameras measuring pixel-level temperatures across -20°C to 1000°C at 640×480 and 384×288 resolutions. Programmed synchronized control for Non-Destructive Testing (NDT) welding inspection using dual 50W modulated diode lasers (808–1064 nm) and automated photo-thermal mapping across six VIS-NIR wavelengths (639–980 nm, 1–10W).

- **Semiconductor Wafer, Photovoltaic & Fiber Laser Scribing Systems (SP2L-SWPSMS / SP2L-LSM / SP2L-LEM):**

  Architected low-level C/C++ and Python hardware bindings over Win32 DLLs (MarkEzd.dll) and microsecond DB25 parallel hardware gating to control >5 kW peak-power NIR semiconductor wafer scribing systems (150×150 mm field on Si and III-V substrates). Synchronized motorized X–Y stage kinematics with 1064 / 532 / 355 nm DPSS optical engines for thin-film solar cell P1, P2, and P3 scribing and 20–100W IPG Photonics pulsed fiber laser systems up to 2500 mm/s.

- **Industrial 3D Bioprinter & Additive Manufacturing Control (Bio4D V2 / InkPrinter / Metal3-D):** Designed a deterministic 100Hz closed-loop pneumatic PID controller in bare-metal C++ interfacing with an MPX5700 pressure transducer for precision bio-ink extrusion. Built dual-MCU lockstep synchronization with a PySide6 + PyOpenGL 3D digital twin supervisory interface using custom M118 serial handshakes.

## KEY ENGINEERING REPOSITORIES & ARCHITECTURE

- **InkPrinter (Bio4D V2 Control Suite):** Dual-MCU lockstep firmware, 100Hz pneumatic PID loop, and PySide6/PyOpenGL 3D bioprinting supervisory interface (github.com/AtharvaYadav-dev/InkPrinter).
- **Galvo_Scriber (Laser Kinematics & Gating Engine):** Custom 2D/3D CAD vector transformations, MarkEzd.dll C++/Python bindings, and microsecond DB25 laser gating (github.com/AtharvaYadav-dev/Galvo_Scriber).
- **Metal3-D (Additive Manufacturing Supervisory Platform):** G-code toolpath parser, real-time machine state diagnostics, and automated execution suite (github.com/AtharvaYadav-dev/Metal3-D).

## EDUCATION

- **Master of Computer Applications (MCA) — Artificial Intelligence & Machine Learning:** Dr. D. Y. Patil Vidyapeeth | 2026 – 2028
- **Bachelor of Science in Information Technology (B.Sc. IT):** Thakur College of Science & Commerce, Mumbai | Graduated May 2026